

ARQUITETURAS DE COMPUTADORES PARA SISTEMAS CRÍTICOS (CCSAR)

Survey Project

Deadline: November 22nd, 2026

Weight: 40% of the final grade

Objective

The aim of this project is to conduct and present high-quality research that explores state-of-the-art techniques discussed in class or presented at leading computer architecture and systems conferences. The deliverable will be a research survey paper that provides a thorough overview of a selected research area, supported by an extensive review of the literature and a critical analysis of existing approaches.

The survey should not simply summarize prior work. Students are expected to identify key research questions, compare competing approaches, discuss architectural trade-offs, highlight open challenges, and evaluate the strengths and weaknesses of the proposed solutions. Where appropriate, comparative tables should be used to summarize the main characteristics, advantages, limitations, and trade-offs of competing approaches.

Students may use generative AI tools as supporting tools, but they remain responsible for the accuracy, originality, and integrity of the submitted work. AI-generated content and references must be verified against the original sources.

This project provides an opportunity to develop expertise in modern computer architectures and systems, making it particularly valuable for students interested in research, innovation, or doctoral studies.

Paper Requirements

Your survey paper should:

1. Summarize the state of the art in a specific research area, citing relevant literature.
2. Present multiple perspectives and competing approaches with appropriate references.
3. Identify the key research questions addressed by the literature.
4. Compare architectural trade-offs, limitations, and design choices.
5. Discuss open challenges and future research directions.
6. Conclude with a critical synthesis of the strengths, weaknesses, and trade-offs of the surveyed approaches.
7. Be structured and formatted as a research paper with proper citations.

8. Include at least 20 relevant references, with most published within the last five years. References should primarily consist of peer-reviewed conference and journal papers.
9. Have an expected length of approximately 8–12 pages (excluding references). The page limit excludes references but includes figures and tables.

Teamwork

The project should be completed in pairs. Each team member must contribute equally to the work. The research findings will be presented orally. All team members must participate in the presentation and should be prepared to answer questions about the work.

Presentation

12–15 minutes, followed by questions.

Suggested Conferences

Architecture-focused conferences:

- ISCA (International Symposium on Computer Architecture)
- MICRO (International Symposium on Microarchitecture)
- HPCA (High-Performance Computer Architecture)
- ASPLOS (Architectural Support for Programming Languages and Operating Systems)
- DSN (Dependable Systems and Networks)
- ISLPED (International Symposium on Low Power Electronics and Design)
- Hot Chips

Systems-focused conferences:

- SOSP (Symposium on Operating Systems Principles)
- OSDI (Operating Systems Design and Implementation)
- EuroSys
- USENIX ATC
- FAST (Conference on File and Storage Technologies)

Suggested Topics

Students are encouraged to select a topic related to their research interests. Topics should be related to computer architecture and, where applicable, connected to the requirements of critical systems, such as performance, efficiency, dependability, security, or predictability. Students should narrow broad topics to a specific research question or architectural problem.

The following topics are provided as examples.

A. Computer Architecture Fundamentals

1. Architectures for Datacenters
 - Power management
 - Datacenter workloads

- Warehouse-scale computing
- Specialized datacenter architectures
- 2. Caching and Memory Systems
 - Replacement policies
 - Prefetching techniques
 - Dead-block prediction
 - DRAM caches
 - High-bandwidth memory (HBM)
- 3. Hardware Support for Virtual Memory
 - Address translation mechanisms
 - Translation lookaside buffers
 - Virtual memory scalability
- 4. Cache Coherence and Multiprocessors
 - Coherence protocols
 - Validation techniques
 - Scalability challenges
- 5. Reliability and Dependability
 - Hard and soft faults
 - Aging and wear-out mechanisms
 - Fault-tolerant architectures
- 6. Interconnection Networks
 - Networks-on-Chip (NoCs)
 - Chiplet interconnects
 - Photonic interconnects
 - Emerging communication technologies

B. Emerging Architectures

- 7. Chiplet-Based Architectures
 - UCIe ecosystem
 - Advanced packaging
 - Heterogeneous integration
- 8. 3D Integration and Stacking
 - 3D memory systems
 - Thermal challenges
 - Performance implications
- 9. Near-Memory and Processing-in-Memory Architectures
 - Data-centric computing
 - Memory-centric architectures
 - In-memory acceleration
- 10. RISC-V Ecosystem
 - Architectural extensions
 - Custom accelerators
 - Industrial adoption
- 11. Neuromorphic Computing
 - Brain-inspired architectures
 - Event-driven computing
 - Low-power processing

12. Quantum Computing Architectures

- Quantum processors
- Qubit technologies
- Interconnect architectures

C. AI and Accelerators

13. AI Acceleration Architectures

- TPUs
- NPUs
- Transformer accelerators
- Large Language Model (LLM) infrastructure

14. Dataflow and Spatial Architectures

- Dataflow execution models
- AI accelerators
- Spatial computing architectures

15. Heterogeneous Computing

- CPU-GPU systems
- FPGA acceleration
- Domain-specific accelerators

16. AI-Assisted Computer Architecture

- Machine-learning-based branch prediction
- Intelligent cache management
- Architecture design-space exploration

D. Security and Sustainability

17. Secure Processor Architectures

- Trusted execution environments
- Confidential computing
- Side-channel attack mitigation

18. Energy-Efficient and Sustainable Computing

- Energy-aware architectures
- Carbon-efficient computing
- Green datacenters
- Energy-proportional systems

Evaluation Criteria

The project will be assessed according to the following criteria:

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| • Quality, breadth, and depth of literature review: | 30% |
| • Critical analysis, comparison, and discussion: | 30% |
| • Organization, structure, and writing quality: | 15% |
| • Relevance and quality of references: | 10% |
| • Oral presentation: | 15% |

Conclusion

This research project is an opportunity to deepen your understanding of the architectural techniques that underpin modern computing systems. Your survey should go beyond summarizing existing work by critically analyzing architectural trade-offs, identifying open challenges, and discussing future directions.

Choose a topic that aligns with your interests and career goals, and ensure that your work is grounded in high-quality, recent research from leading conferences and journals. For topics related to critical systems, students must explicitly discuss how architectural choices affect relevant system properties such as performance, efficiency, dependability, security, or predictability.